

Brownian Regularity of the KPZ fixed point

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Cambridge probability seminar

Joint work with Sourav Sarkar

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Overview

- 1 Random interface growth in $(1 + 1)$ dimensions
- 2 The KPZ fixed point and the universality class
- 3 “Brownianness” of the KPZ fixed point
- 4 Proofs

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In 1986, Kardar, Parisi and Zhang [[Kardar et al., 1986](#)] predicted that many planar random growth processes possess universal scaling behaviour, with local dynamics governed by the KPZ equation:

$$\partial_t h = \underbrace{(\partial_x h)^2}_{\text{lateral growth}} + \underbrace{\partial_x^2 h}_{\text{smoothing}} + \underbrace{\xi}_{\text{spacetime white noise}} \underbrace{h(t, x)}_{\text{height at time } t, \text{ position } x}$$

Eden model

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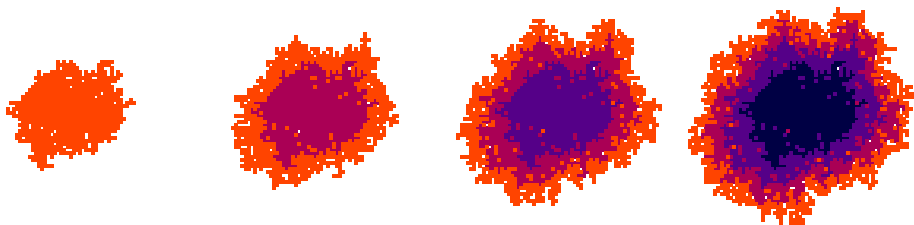


Figure: Simulation of Eden growth model at four different epochs, adding 800 new blocks per epoch.

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- **Scaling limit:** $c_2 T^{-1/3}(h(T, 0) - c_1 T)$ converges to *Tracy-Widom* distribution from random matrix theory.

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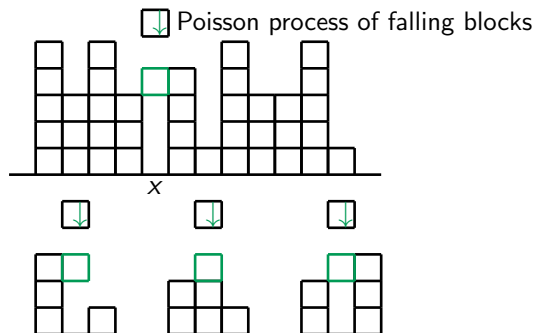


Figure: Ballistic deposition model. Growth occurs (here at site x) when blocks stick to first point of contact (denoted in green)².

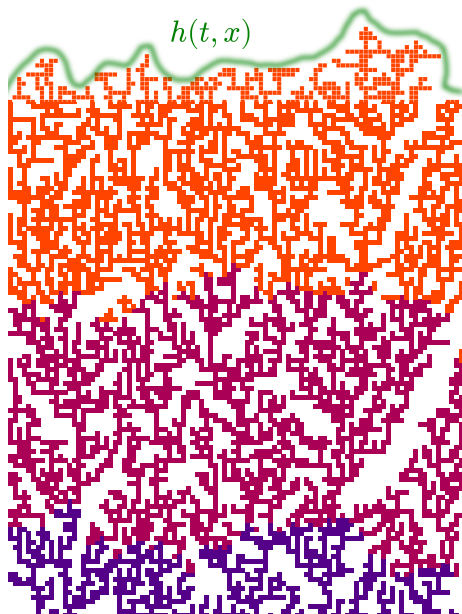
²Inspired by Figure 0b in [\[Corwin, 2011\]](#).

Ballistic deposition run for long time

- Full scaling limit:

$$\varepsilon^{1/2} h(\varepsilon^{-3/2} t, \varepsilon^{-1} x) - C_\varepsilon t$$

as a process in x for fixed $t > 0$,
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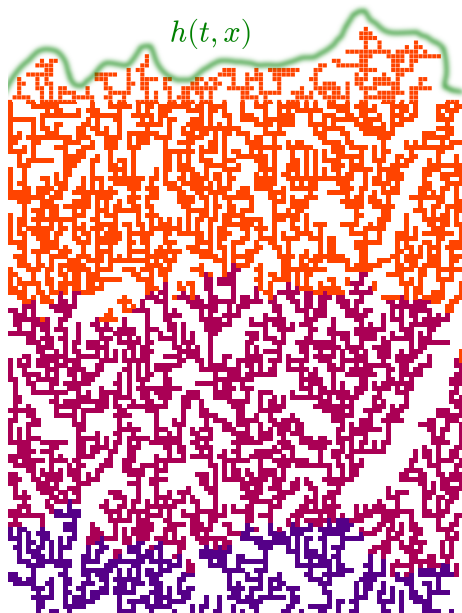
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- Fluctuation : Space : Time =
1 : 2 : 3



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- All models in this class have a height function which has the “same asymptotic fluctuations” under the 1:2:3 scaling as the KPZ equation.

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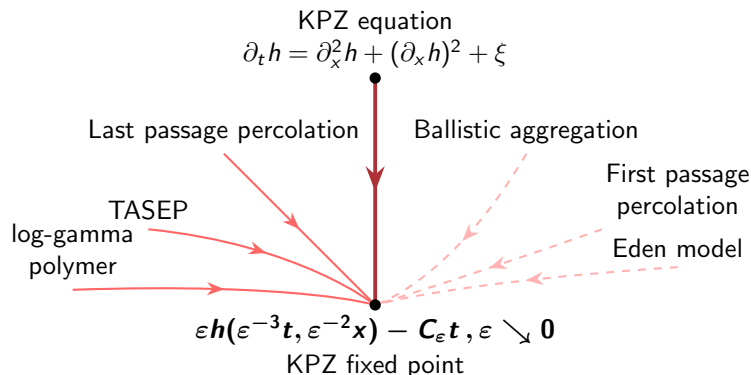
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- Analogue of the **Donsker's theorem** in this universality class!
- *A model is in the KPZ universality class, if starting from **any height**, its height under 1 : 2 : 3 scaling converges to the KPZ fixed point.*

What is the KPZ universality class?



Not exactly solvable

KPZ fixed point for special initial conditions

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- Does it still look “locally like a Brownian motion”?

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Definitions: local Brownianness

Local Brownian limit

F has local Brownian limit if the scaled process

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F is called Brownian on compact if for all $y_1 < y_2$, the law of $F(\cdot) - F(y_1)$ on $[y_1, y_2]$ is absolutely continuous wrt that of a Brownian motion (with diffusion parameter 2) starting from $(y_1, 0)$ on $[y_1, y_2]$.

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- Brownian on compact $\implies$ local Brownian limit.

Which is which?

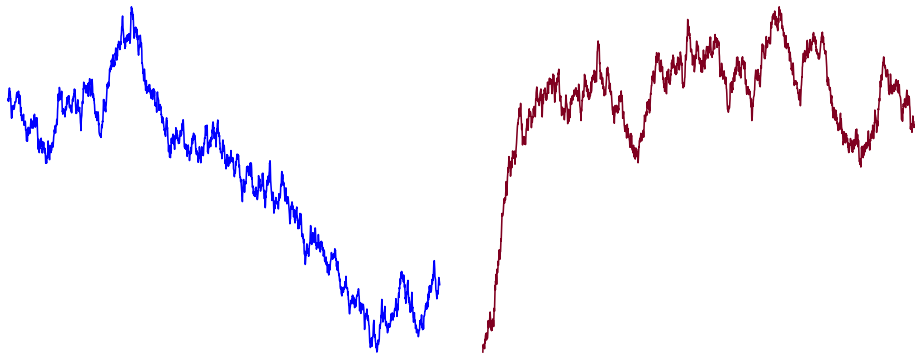


Figure: Which is which? One is locally Brownian, one is not.

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- The name comes from the fact that $h_t(x)$ is finite for all $x \in \mathbb{R}$ if and only if the initial condition f is in $\mathcal{I}_t$.
 - For example, a function f satisfying $f(x) \leq C + x^2/t - |x| \log(1 + |x|)$ for some $C > 0$, is in $\mathcal{I}_t$.
 - Essentially want **sub-parabolic growth**.

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- [Dauvergne, 2024]: L^∞ bound on Radon-Nikodym derivative of Airy_2 (up to height shift) on compacts.

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Application:

- KPZ fixed point has a **unique maxima on compacts** (Johansson's conjecture!). In particular, Airy_1 process has a unique maxima on compacts.

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Conjecture, [Calvert et al., 2019]

The Radon-Nikodym derivative of the KPZ fixed point with arbitrary initial data with respect to Brownian motion is in L^p for all $p > 1$!

Main result

Quantitative Brownian regularity, [Tassopoulos and Sarkar, 2025a]

The law of the KPZ fixed point started from arbitrary (finitary) initial data on some fixed interval, ν , exhibits quantitative Brownian regularity of the form

$$\nu(A) \leq c \exp \left(-d \log^r \log (1/\mu(A)) \right) ,$$

for some positive constants $c, d > 0, r \in (0, 1)$ and all events A , where μ denotes the Wiener measure.

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KPZ fixed point geometry: the Airy line ensemble and the Airy Sheet

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- The proof uses a geometric representation of the KPZ fixed point as a variational formula in terms of a random metric, also known as the the Airy sheet.
- The **Airy sheet** was constructed by [\[Dauvergne et al., 2022\]](#). The standard Airy sheet $\mathcal{S} : \mathbb{R}^2 \mapsto \mathbb{R}$ is a random continuous function defined as a scaling limit of Brownian last passage percolation.

Detour: Last passage percolation

- For a sequence of continuous functions $f = (f_1, f_2, \dots)$ and $x \leq y$ and $n \leq m$, we define a **path** from (x, ℓ) to (y, m) as a non-increasing function $\pi : [x, y] \mapsto \mathbb{N}$ with $\pi(x) = \ell$ and $\pi(y) = m$. We define the **length** of π as the **sum of increments of f along the path π** .

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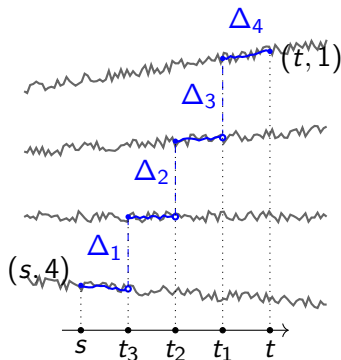


Figure: length $\ell = \sum_{i=1}^4 \Delta_i$.

Last passage percolation contd...

- For $x \leq y \in \mathbb{R}$ and $m < \ell \in \mathbb{N}$, we define the **last passage value** of f from (x, ℓ) to (y, m) by

$$f[(x, \ell) \rightarrow (y, m)] := \sup_{\pi} \ell(\pi),$$

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- When f_i 's are independent random walks with exponential steps (DLPP) or independent Brownian motions (**Brownian LPP**), they are in the KPZ universality class.

Airy sheet scaling limit

With $(x, s)_n = (s + 2x/n^{1/3}, -\lfloor sn \rfloor)$, the translation between limiting and pre-limiting locations, we have the following (Theorem 8.3 [Dauvergne et al., 2022]).

Theorem (Scaling limit of Brownian last passage percolation)

There exists a coupling of Brownian last passage percolation and the Airy sheet $\mathcal{S}$ so that

$$B_n[(x, 0)_n \rightarrow (y, 1)_n] = 2\sqrt{n} + 2(y - x)n^{1/6} + n^{-1/6}(\mathcal{S} + o_n)(x, y).$$

Here each $B_n \in \mathcal{C}^{\mathbb{Z}}$ is an ensemble of independent two-sided Brownian motions. Each o_n is a random function asymptotically small in the sense that on every compact set $K \subset \mathbb{R}^2$ there exists a $\alpha > 1$ with $\mathbb{E} a^{\sup_K |o_n|^{3/4}} \rightarrow 1$.

Airy sheet profile

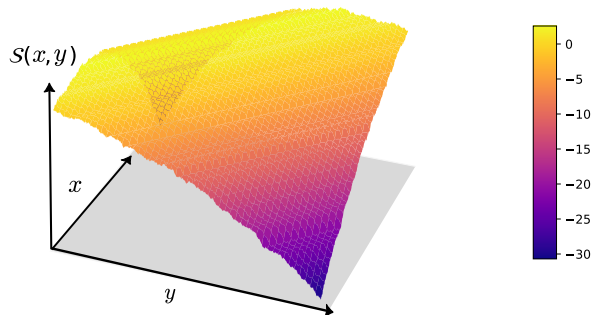


Figure: Simulation of the Airy sheet on $[-3, 3] \times [-3, 3]$ using approximation by Brownian last passage values. Note the parabolic curvature.

KPZ fixed point as a variational formula

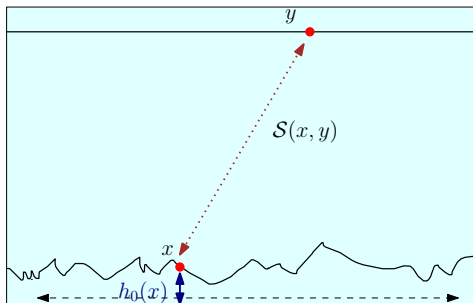
- The KPZ fixed point at time 1 can be written in terms of a random metric, the Airy sheet $\mathcal{S}$ and the initial condition $h_0 : \mathbb{R} \mapsto \mathbb{R} \cup \{-\infty\}$ as

$$h_1(y) = \sup_{x \in \mathbb{R}} \left(h_0(x) + \mathcal{S}(x, y) \right), \quad y \in \mathbb{R}.$$

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Airy sheet coupling

Definition (Airy Sheet)

The **Airy Sheet** is a random continuous function $\mathcal{S} : \mathbb{R}^2 \rightarrow \mathbb{R}$ so that the following hold

- ① Stationarity: $\mathcal{S}$ has the same law as $\mathcal{S}(\cdot + t, \cdot + t)$ for all $t \in \mathbb{R}$
- ② Coupling: $\mathcal{S}$ can be coupled with a parabolic Airy line ensemble so that $\mathcal{S}(0, \cdot) = \mathcal{A}_1(\cdot)$ and almost surely for all triplets $(x, y, z) \in (0, \infty) \times \mathbb{R}^2$ there exist random integers $K_{x,y,z}$ such that for all $k \geq K_{x,y,z}$

$$\mathcal{A}[(-\sqrt{k/(2x)}, k) \rightarrow (z, 1)] - \mathcal{A}[(-\sqrt{k/(2x)}, k) \rightarrow (y, 1)] \quad (1)$$

$$= \mathcal{S}(x, z) - \mathcal{S}(x, y). \quad (2)$$

The Airy line ensemble

Theorem (Theorem 2.1 [Dauvergne et al., 2022])

Let WB^n be a Brownian n -melon. Define the rescaled melon $A^n = (A_1^n, \dots, A_n^n)$ by

$$A_i^n(y) = n^{1/6} \left((WB^n)_i(1 + 2yn^{-1/3}) - 2\sqrt{n} - 2yn^{1/6} \right).$$

Then A^n converges to a random sequence of functions $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \dots) \in C^{\mathbb{N}}$ in law with respect to product of uniform-on-compact topology on $C^{\mathbb{N}}$. For every $i < j$, we have $\mathcal{A}_i(y) > \mathcal{A}_j(y)$ for all $y \in \mathbb{R}$ almost surely. The function $\mathcal{A}$ is called the **(parabolic) Airy line ensemble**.

Pitman transform

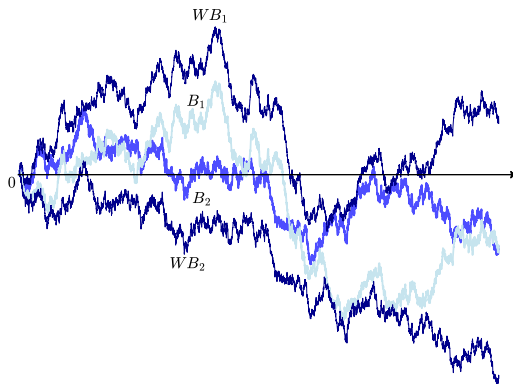


Figure: An illustration of the Pitman transform WB of two Brownian motions B_1, B_2 .

The Airy line ensemble

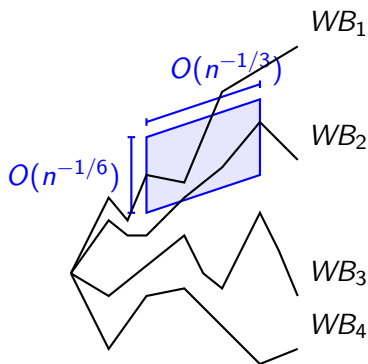


Figure: Brownian Melon Scaling limit. Above is a realisation of of the $WB^n, n = 4$ melon. “Zooming in ” on the parallelogram with at small scales and taking the limit as $n \rightarrow \infty$ yields the convergence in law to the (parabolic) Airy Line Ensemble.

Fixed point reduction

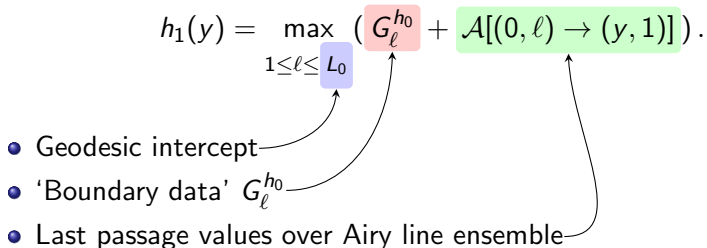
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$$h_1(y) = \max_{1 \leq \ell \leq L_0} (G_\ell^{h_0} + \mathcal{A}[(0, \ell) \rightarrow (y, 1)]).$$


- Geodesic intercept
- ‘Boundary data’ $G_\ell^{h_0}$
- Last passage values over Airy line ensemble

Airy line ensemble last passage values

To obtain control over $\mathcal{A}[(0, \ell) \rightarrow (y, 1)]$, $y \geq 0, \ell \geq 1$ we use the **Brownian Gibbs resampling property** of the Airy Line ensemble [Corwin and Hammond, 2014], which briefly states that for $a < b$, $k \in \mathbb{N}$, the law of the Airy line ensemble restricted to the region

$$\{1, 2, \dots, k\} \times (a, b),$$

$\mathcal{A}|_{\{1, 2, \dots, k\} \times (a, b)}$, conditionally on all the data generated by the Airy line ensemble outside of this region,

$$\{A_i(x) : (i, x) \notin \llbracket 1, k \rrbracket \times (a, b)\},$$

is given by k independent Brownian bridges with entry data $\underline{x} = (\mathcal{A}_i(a))_{1 \leq i \leq k}$, $\underline{y} = (\mathcal{A}_i(b))_{1 \leq i \leq k}$ conditioned to not intersect and stay above $f = \mathcal{A}_{k+1}$ on (a, b) .

Brownian Gibbs

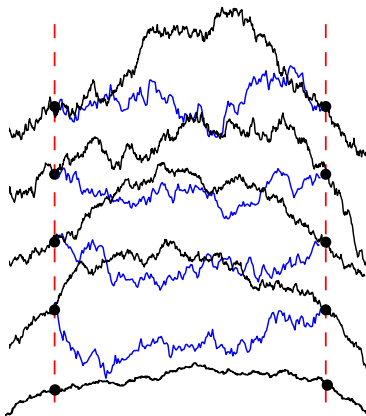


Figure: Figure illustrating the Brownian Gibbs property on the first four lines of the parabolic Airy Line ensemble $\mathcal{A} = \{\mathcal{A}_1 > \mathcal{A}_2 > \dots\}$ (in **black**) between two points (indicated by the **red** vertical dashed lines). The **blue** curves represent resampled versions of first four lines in the ensemble between the endpoints, conditioning on everything else and avoiding the fifth line.

Reduction to inhomogeneous Brownian LPP

The above bounds allow us with high probability to reduce the problem of Brownian regularity of the KPZ fixed point to that of the **inhomogeneous Brownian last passage values**,

$$\max_{k \leq \ell \leq m} (g_\ell + B[(0, \ell) \rightarrow (t, k)]), \quad t \geq 0$$

for $1 \leq k \leq m$ and $g_1 \geq \dots \geq g_m$. In [\[Sarkar and Virág, 2021\]](#), it was already demonstrated that its top line is absolutely continuous with respect to Brownian motion on compacts. We strengthened this comparison and obtained the optimal estimates.

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Radon-Nikodym derivative estimates for Inhom. BLPP, [Tassopoulos and Sarkar, 2025b]

The Radon Nikodym derivative of the law of $H_m := \max_{1 \leq \ell \leq m} (g_\ell + B[(0, \ell) \rightarrow (t, 1)])$ away from zero against a rate two Brownian motion on compacts is in $L^{\infty-}$. In particular, the L^p norms, for $p > 1$, grow like $\mathcal{O}_p(e^{d_p m^2 \log m})$, for some $d_p > 0$.

Inhomogeneous Brownian LPP

The estimates on the Radon-Nikodym derivatives of inhomogeneous BLPP are obtained observing that

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- this allows us to use explicit transition densities of the latter process to obtain bounds on the Radon-Nikodym derivatives, by relating the inhomogeneous case to the homogeneous one
- and use the fact that the top line of homogeneous BLPP is actually Dyson Brownian motion (through a continuous analogue of the RSK correspondence, [\[O’Connell and Yor, 2002\]](#)).

Boundary data

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- Boundary data $G_\ell^{h_0}$, $\ell \geq 1$ can be estimated for some $x_0 > 0$ in the max-plus support of initial data h_0 , K by
 - ▶ pointwise bounds on the initial data on K
 - ▶ pointwise bounds on $G_1^{h_0}$ which is coupled to the Airy sheet
 - ▶ $\lim_{\substack{k \rightarrow \infty \\ x > 0}} |\mathcal{A}[(-\sqrt{k/(2x)}, k) \rightarrow (0, \ell)] - \mathcal{A}[(-\sqrt{k/(2x)}, k) \rightarrow (0, 1)]|,$

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- Theorem 3.7 in [Sarkar and Virág, 2021] shows the limit a.s. stabilises eventually for $k \geq K_x$, (due to the parabolic ‘curvature’ of the Airy line ensemble). In [Tassopoulos and Sarkar, 2025a], we obtain explicit tails for such ‘coalescence’ depths K_x , keeping track of dependence on ℓ . In particular, we have for any $\theta > 0$,

$$\inf \{m \geq 1 : \mathbb{P}(K_x \geq m) \leq \theta\} \leq C\ell^{256} (\exp(d \log^{1000}(1/\theta)))$$

for some $C, d > 0$.

Boundary data

- The non-intersecting nature of the Airy line ensemble induces an effective ‘truncation’ on the boundary data differences, bounding them in terms of the Airy line ensemble differences.
- This gives a uniform comparison for increments of the KPZ fixed point (independent of h_0).

Geodesic transversal fluctuations

We obtain ‘stretched exponential’ concentration of geodesic intercepts L_0 ,

Theorem, [Tassopoulos and Sarkar, 2025a]

$$\sup_{k \in \mathbb{N}} \exp(dk^{1/126}) \cdot \mathbb{P}(L_0 \geq k) < \infty,$$

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- This was achieved by establishing quantitative control over jump times of last passage geodesics in the prelimiting (Brownian melon) environments converging to the Airy line ensemble (under edge-scaling).

Transversal fluctuations of semi-infinite geodesics

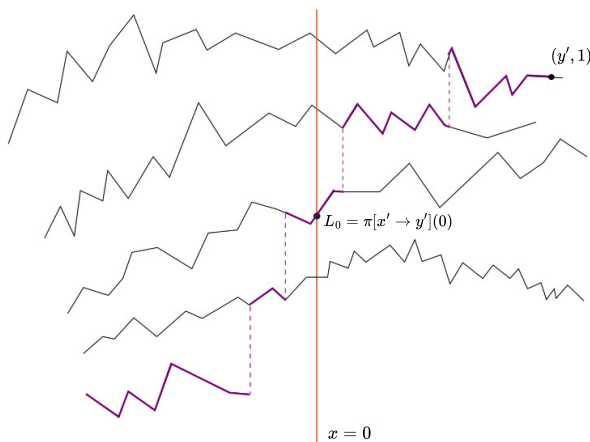


Figure: Above is displayed the point $(0, L_0)$ at which the last passage path $\pi[x' \rightarrow y']$ on the Airy line ensemble $\mathcal{A} = (\mathcal{A}_1, \mathcal{A}_2, \dots)$ (purple) meets with the axis $\{x = 0\}$, where $x', y' > 0$. Here $L_0 = 3$ and the first four lines of $\mathcal{A}$ are shown.

Applications of Quantitative Brownian regularity

- One can obtain a one-sided large deviation inequality for spatial increments of the KPZ fixed point on compacts.

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$$\mathbb{E}[X \cdot \mathbf{1}_B] \leq f(\mathbb{P}(X \in B)) \leq c e^{-d \log^r \log 1/\mathbb{P}(X \in B)}, \quad B \subseteq (0, \infty) \text{ Borel}.$$

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- The above improvement on the tails can also be converted into an integrability statement for the Radon-Nikodym derivative.

One-sided large dev ineq.

Corollary

Fix $a > 0$ and a bounded set $I \subseteq \mathbb{R}$. Consider the set

$$E_a \stackrel{\text{def}}{=} \{\omega \in H : I(\omega) \geq a\}, \quad I(\omega) \stackrel{\text{def}}{=} \int_{[0, \sup I - \inf I]} |\dot{\omega}(t)|^2 dt, \omega \in H$$

where we denote the Cameron-Martin space by H . Then, there exist $0 < r < 1$ and $C > 0$ such that for all F closed and initial data $h_0 : \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$, the laws $\nu_\varepsilon^{h_0}$ of $\sqrt{\varepsilon}(h_1(\cdot + \inf I) - h_1(\inf I))$, satisfy

$$\begin{aligned} \limsup_{\varepsilon \rightarrow 0} \sup_{h_0 \text{ } t\text{-finitary}} \log \nu_\varepsilon^{h_0}(F) + C \log^r \left(\frac{a}{\varepsilon} \right) \\ \leq -C \log^r \left(\frac{\inf_{\omega \in F} I(\omega)}{a} \right), \quad F \cap E_a \neq \emptyset. \end{aligned}$$

Applications contd.

Corollary

Fix arbitrary initial data $h_0 : \mathbb{R} \rightarrow \mathbb{R} \cup \{-\infty\}$. Let X^{h_0} denote the Radon-Nikodym derivative of the law of the increments of the KPZ fixed point started from h_0 , against the rate two Wiener measure μ . Then there exists a function satisfying

$$\Phi(0) = 0, \quad \lim_{x \rightarrow \infty} \frac{\Phi(x)}{x} = \infty,$$

such that

$$\mathbb{E}[\Phi(X^{h_0})] < \infty.$$

In particular, we have the lower bounds

$$\Phi(x) \geq \left[e^{d \log^r \log x} \right] \cdot x, \quad x > x_0,$$

for some $x_0 > 1$ sufficiently large and constants $d, r > 0$.

Future directions!

- Can we prove the Radon-Nikodym derivative is in L^p for all $p > 1$?

Future directions!

Ideally want some $p > 1$, $d \geq 1$, $r > 0$ such that

- we have the estimate for the finite depth truncations H_m , $m \geq 1$ as defined in slide 35 and A Borel (depending only on increments on some bounded set),

$$\mathbb{P}(H_m(\cdot) \in A) \leq c_p e^{m^d} \cdot \mu(A)^{1-1/p}, \quad m \geq 1,$$

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for the same $r > 0$.

- We believe that one can achieve $r = 3$ from transversal fluctuation of geodesics in discrete environments. Moreover, we also expect to have $d < 3$ from our already established inhomogeneous Brownian LPP estimates. This would give $r > d$, as required for L^{p-} -regularity.

Future directions!

- Can we distinguish the KPZ fixed points started from different initial data?

Thank you for your attention!

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